AP PRECALCULUS

Unit 1 • Worksheet 1.12A • Answer key

Worksheet 1.12A • Transformations of Functions

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

How is the graph of $g(x) = f(x+3) - 1$ obtained from the graph of f?

(A) right 3, down 1
(B) left 3, down 1
(C) left 3, up 1
(D) right 3, up 1

Solution

Answer: (B).

Question 2

NO CALCULATOR

How is the graph of $g(x) = f(2x)$ obtained from the graph of f?

(A) vertical dilation by 2
(B) horizontal dilation by 2
(C) horizontal dilation by $\frac{1}{2}$
(D) vertical dilation by $\frac{1}{2}$

Solution

Answer: (C).

Question 3

NO CALCULATOR

How is the graph of $y = -f(x)$ obtained from the graph of f?

(A) reflection over the x-axis
(B) reflection over the y-axis
(C) reflection over $y = x$
(D) shift down 1

Solution

Answer: (A).

Question 4

NO CALCULATOR

Which describes $g(x) = 3f(-x)$?

(A) reflect over the x-axis, vertical dilation by 3
(B) reflect over the y-axis, vertical dilation by 3
(C) reflect over the y-axis, horizontal dilation by 3
(D) shift left 3

Solution
Answer: (B).

Question 5 (Free response)

NO CALCULATOR

Describe, in order, the transformations that take the graph of $f(x) = x^2$ to the graph of each function.
(a) $g(x) = (x-4)^2 + 1$ (b) $h(x) = -2(x+1)^2$ (c) $k(x) = \frac{1}{2}(x-3)^2 - 5$

Solution
(a) Right 4, up 1. (b) Left 1, vertical dilation by 2, reflection over the x-axis. (c) Right 3, vertical dilation by $\frac{1}{2}$, down 5.